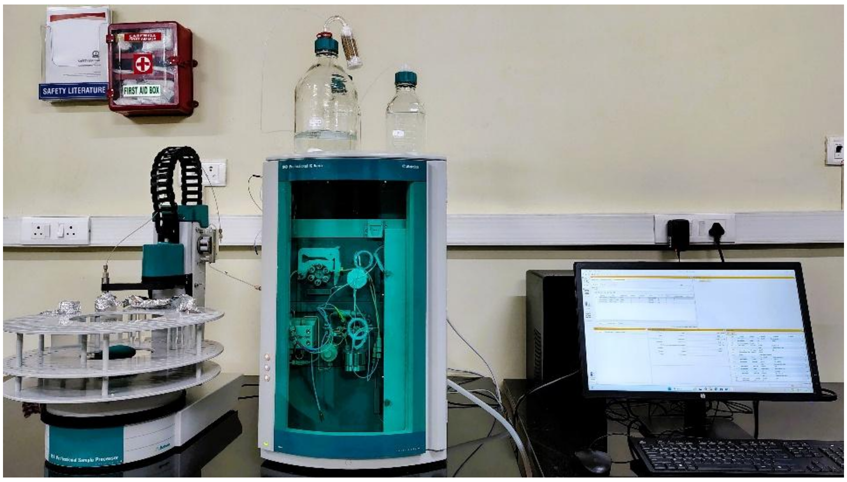
Anions & Cations Analysis @ ICWaR



Our laboratory offers high-performance Ion Chromatography (IC) services using the **Metrohm 940 Professional IC Vario,** designed for sensitive, reliable trace-level ion analysis. This system enables the separation and quantification of inorganic anions, cations, organic acids, and other ionic species in water, environmental, and industrial samples. Ion chromatography is widely used for monitoring water quality, industrial effluents, and environmental contaminants, where accurate ionic profiling is essential for quality control and regulatory compliance.

Instrument Capabilities

- **Intelligent Components** (iPump, iDetector, iColumn & MagIC Net Software) - automated control, simplified method development, and reliable data acquisition.
- **High-Pressure Pump** - precise solvent delivery up to 50 MPa for reproducible separations.
- **Conductivity Detection** - low baseline noise with high measurement stability.
- **MSM Chemical Suppressor** - enhanced sensitivity and reduced background for accurate anion determination.
- **Column Thermostat** - temperature control up to 80°C for stable retention times.
- **Autosampler** - automated sequences, repeat injections, calibration runs, and high-throughput operation.

Key Advantages of IC Analysis

✓ **Multi-ion analysis** with dedicated flow paths for anion or cation modes.
✓ **High sensitivity & reproducibility** via anion suppressor-enhanced conductivity detection.
✓ **Reliable in complex matrices** through inline filtration, degassing, and pulse dampening.
✓ **Automated calibration & reporting** via MagIC Net software.
✓ **System suitability testing** for linearity, noise, drift, and injector reproducibility.

Ion chromatography analysis services are ideal for drinking water quality assessment, wastewater and industrial effluent monitoring, environmental studies, and regulatory compliance testing. Whether routine ion monitoring or specialised trace-level ionic analysis is required, this laboratory provides accurate, reliable, and cost-effective analytical solutions.